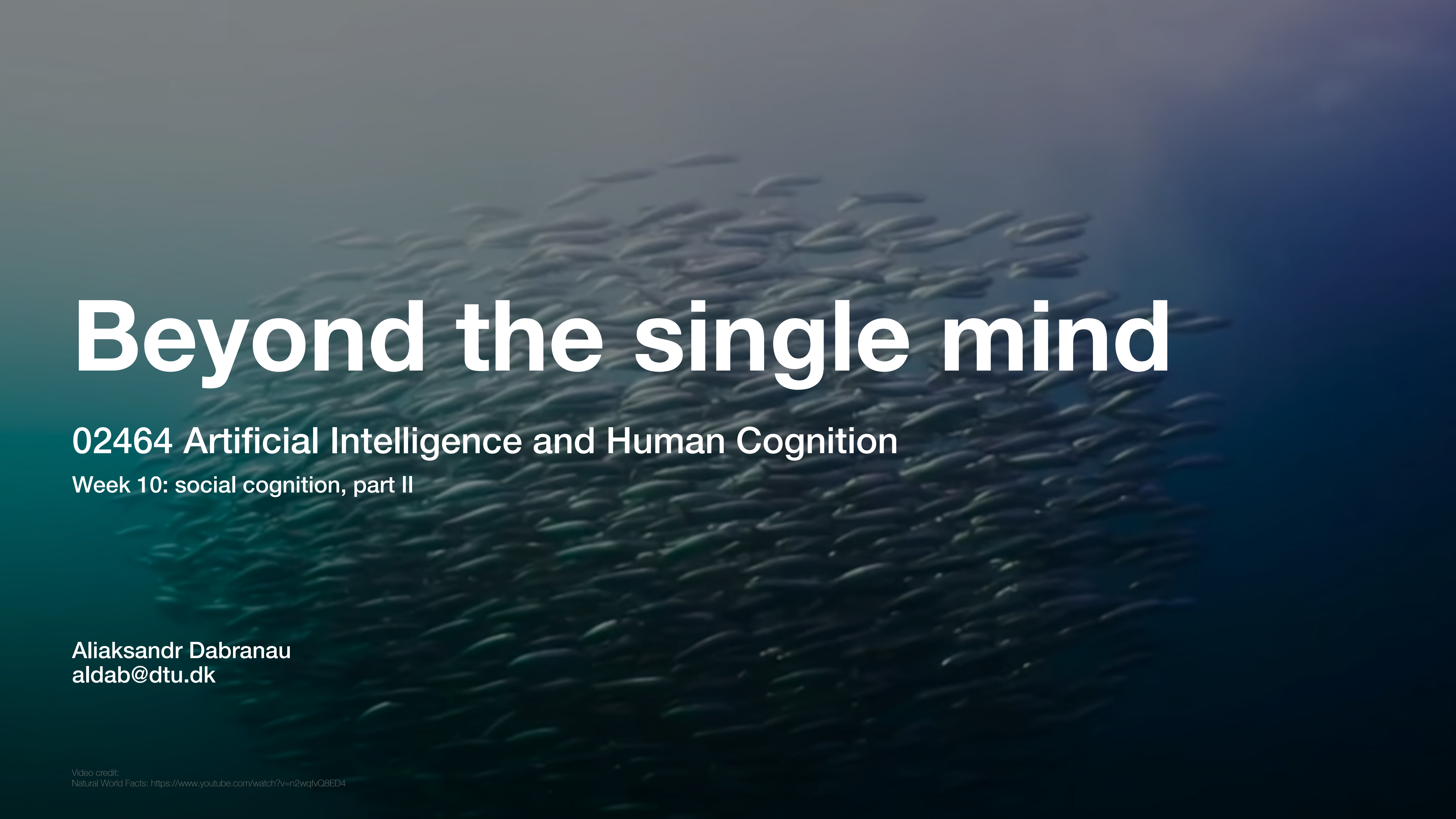


Beyond the single mind

02464 Artificial Intelligence and Human Cognition

Week 10: social cognition, part II

Aliaksandr Dabranau
aldab@dtu.dk

Video credit:
Natural World Facts: https://www.youtube.com/watch?v=n2wqfvQ8ED4

Last week:
Single-mind aspects of social cognition

- Mentalizing and theory of mind
- Mirroring
- Brain substrate used to achieve those functions
- Prejudice

Why “single-mind”?
Because they don’t require actual interaction with the other person.
We are <u>just observers</u> of something external.

Social interaction involves a combination of <u>only single-mind</u> cognitive processes?

Representations

Words:

“Map”

Pictures:

Mental representations:

Components of knowledge in the mind, such as:

- concepts
- rules
- images (e.g., a snapshot of a visual scene in memory)
- …

Thagard, P. (2005). Mind: Introduction to cognitive science. MIT press.

This week:
Group-level aspects of social cognition

Lower-level (not only representational):
- action coordination and joint action
- joint attention

Higher-level (mostly representational):
- group memory
- group decision-making
- conversational alignment

unfold only in actual (direct or mediated) social interaction

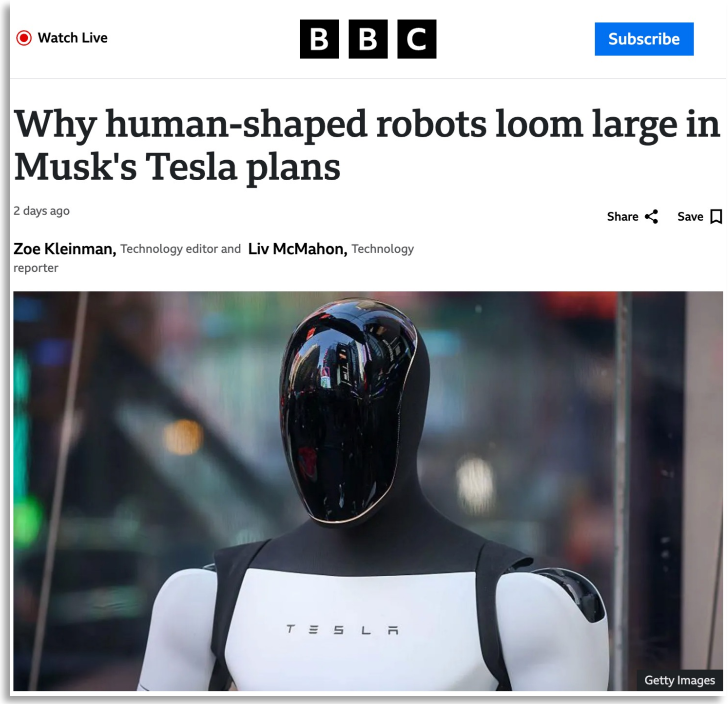
Large-scale (acquire different properties):
- crowd dynamics
- collective attention
- opinion dynamics (e.g., in politics)
- national, cultural, generational memory, etc.

Gu, F., Guiselin, B., Bain, N., Zuriguel, I., & Bartolo, D. (2025). Emergence of collective oscillations in massive human crowds. Nature, 638(8049), 112-119.

Lorenz-Spreen, P., Mønsted, B. M., Hövel, P., & Lehmann, S. (2019). Accelerating dynamics of collective attention. Nature communications, 10(1), 1759.

Why is collective cognition important for AI?

AI has already become (with varying success) an agent participating in group and collective cognition phenomena on different levels — we need it to be better at maximizing humans’ benefit



Watch Live BBC Subscribe

Why human-shaped robots loom large in Musk's Tesla plans

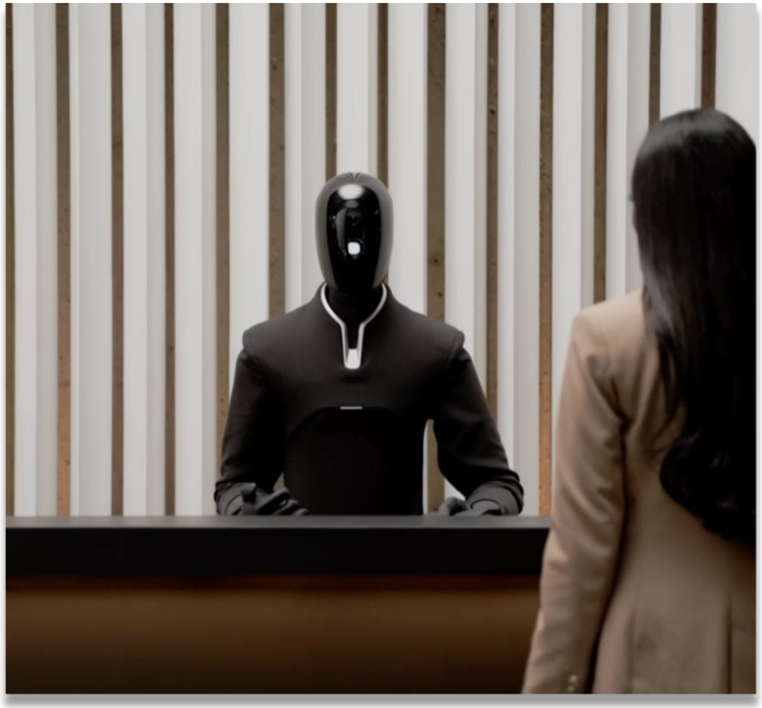
2 days ago

Share Save

Zoe Kleinman, Technology editor and Liv McMahon, Technology reporter

Getty Images

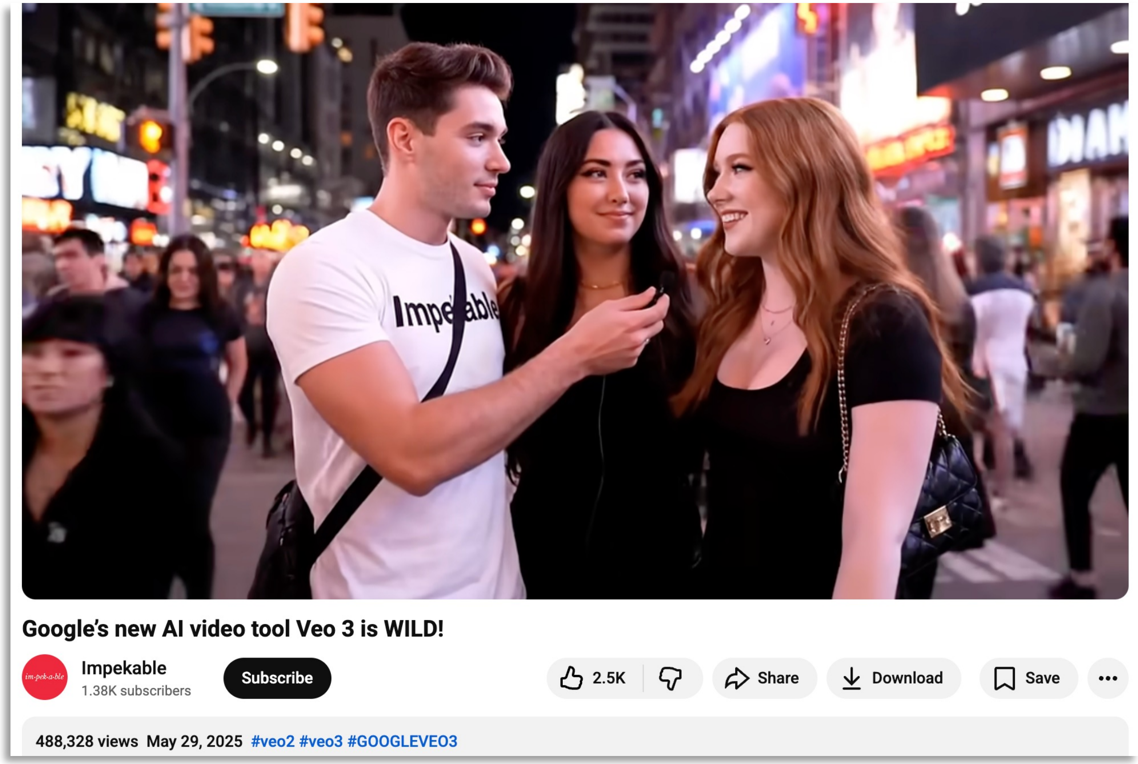
https://www.bbc.com/news/articles/cdrz2rdlykdo



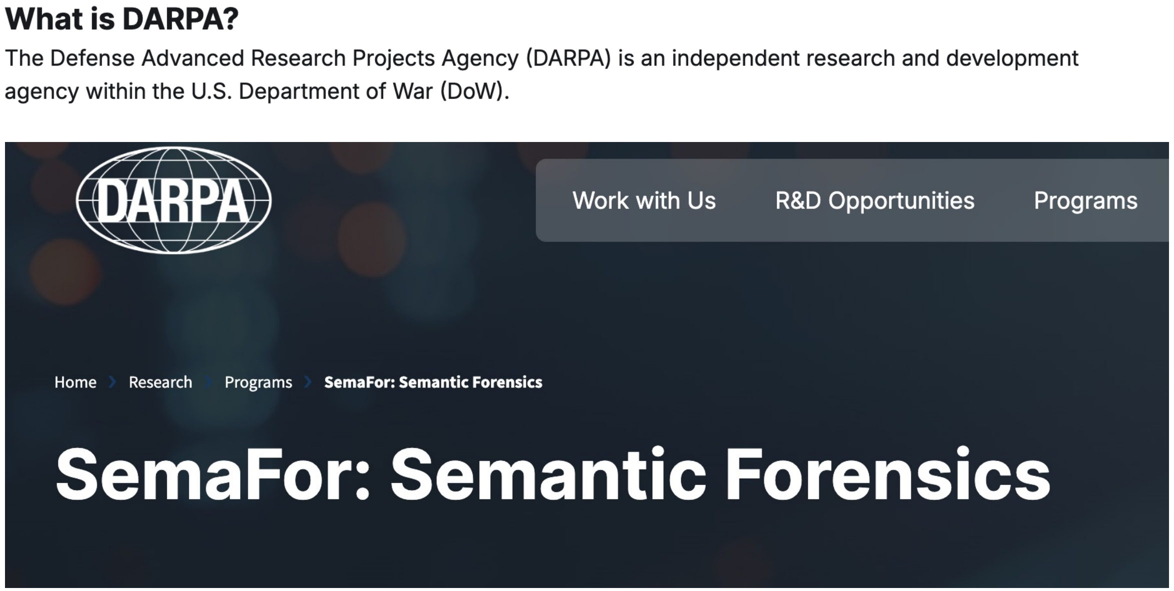
https://www.figure.ai/

Why is collective cognition important for AI?

Development of better tools for the detection of deepfakes



https://www.youtube.com/watch?v=gcZwE5cM4xs



https://www.darpa.mil/research/programs/semantic-forensics

What is special about social interaction?

Does social interaction equal the sum of contributions of interacting people, or is there something additional to it?

The text below addresses this question. The text consists of direct quotations from

De Jaegher, H., & Di Paolo, E. (2007). Participatory sense-making: An enactive approach to social cognition. Phenomenology and the cognitive sciences, 6(4), 485-507.

“Something that is not so common in cases of purely physical coupling, but that we find in the social domain, is that patterns of coordination can directly influence the continuing disposition of the individuals involved to sustain or modify their encounter.”

Interaction dynamic impacts people’s behavior, which shapes interaction dynamics, which impact people’s behavior…

Example: the game of charades — co-constructed representation at one step allows to elaborate meaning at the next step.

“At the same time, it is vital to avoid the error of considering only the interaction and ignoring the individual elements in it.”

Action coordination, joint action, and joint attention

Action coordination

Individual-level behavioral rules may be sufficient for some collective behaviors:

1. Collision avoidance: avoid collisions with nearby flockmates

2. Velocity matching: attempt to match velocity with nearby flockmates

3. Flock centering: attempt to stay close to nearby flockmates

Reynolds, C. W. (1987, August). Flocks, herds and schools: A distributed behavioral model. In Proceedings of the 14th annual conference on Computer graphics and interactive techniques (pp. 25-34).

Photo credit: James Wainscoat, Unsplash

Action coordination

Human coordination is more complex in organization and has more complex goal-structure





Photo credit: unsplash.com

Action coordination and joint action

joint action

= “any form of social interaction whereby two or more individuals coordinate their actions in space and time to bring about a change in the environment” (Sebanz et al., 2006)

action coordination

= “how individuals adjust their actions to those of another person in time and space” (Sebanz et al., 2006)

1. individual and shared representations: goals, plans, task representations, partners’ abilities, etc.

2. processes of monitoring, prediction, and adaptation

3. coordination smoothers — making yourself more predictable: synchronization, speeding up, implicitly negotiating leader-follower roles (can also be at the highest level if predetermined), etc.

Sebanz, N., Bekkering, H., & Knoblich, G. (2006). Joint action: bodies and minds moving together. Trends in cognitive sciences, 10(2), 70-76.

Vesper, C., Butterfill, S., Knoblich, G., & Sebanz, N. (2010). A minimal architecture for joint action. Neural networks, 23(8-9), 998-1003.

Action coordination

intentional
— goal driven, agreed-on, follows the full joint-action structure

unintentional (often interesting on its own)
— spontaneous, with minimal or absent shared representations, reduced monitoring and prediction

Action coordination

Greater synchrony corresponds to more liking

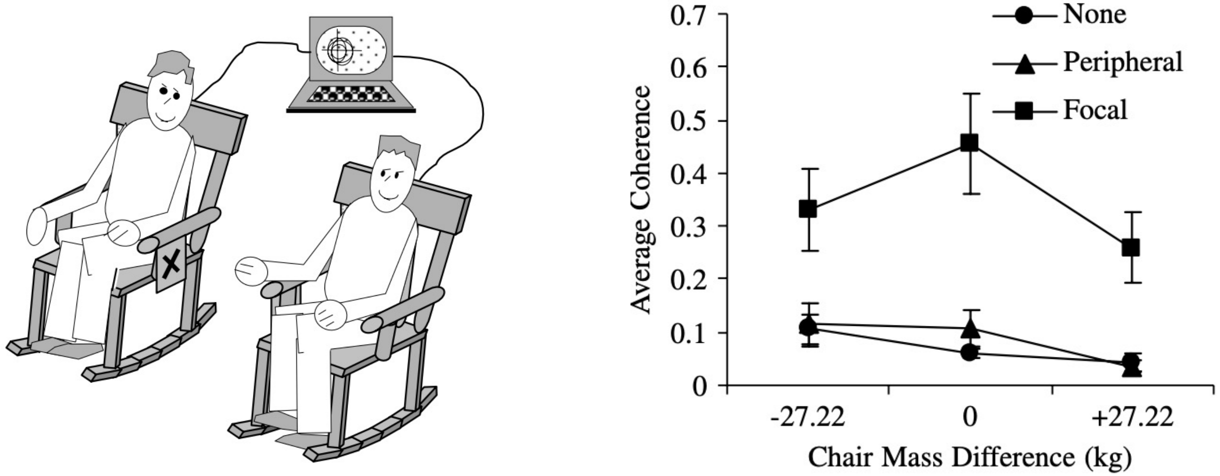
Hove, M. J., & Risen, J. L. (2009). It's all in the timing: Interpersonal synchrony increases affiliation. *Social cognition*, *27*(6), 949-960.

unintentional (often interesting on its own) — spontaneous, with minimal or absent shared representations, reduced monitoring and prediction

Synchronized clapping


Photo credit: Unsplash

More sync even without instructions, via action-perception coupling



Richardson, M. J., Marsh, K. L., Isenhower, R. W., Goodman, J. R., & Schmidt, R. C. (2007). Rocking together: Dynamics of intentional and unintentional interpersonal coordination. Human movement science, 26(6), 867-891.

Action coordination

intentional
— goal driven, agreed-on, follows
the full joint-action structure

Important processes:

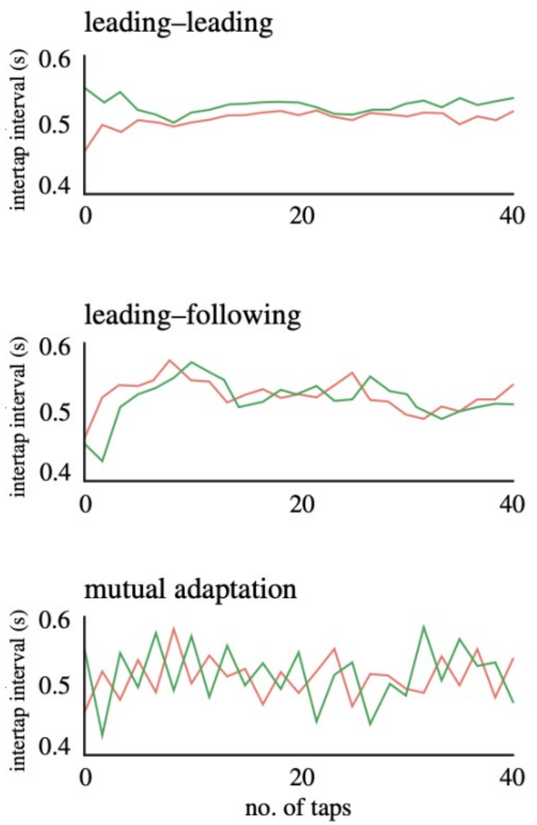
level 2 of joint action structure

Prediction: anticipate each other's actions, and continually adjust their predictions based on the outcome

Adaptation: adjusting to the established rhythm / partner

level 3

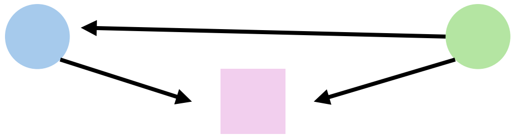
Implicit negotiation of leader-follower roles: leaders are less-adaptive partners and followers are more-adaptive partners



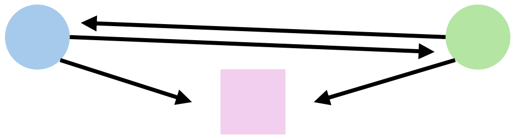
Heggli, O. A., Konvalinka, I., Kringelbach, M. L., & Vuust, P. (2021). A metastable attractor model of self–other integration (MEAMSO) in rhythmic synchronization. Philosophical Transactions of the Royal Society B, 376(1835), 20200332

Joint attention

Shared attention = one of the individuals is aware of herself and the other as being attentive towards the same external entity (=gaze-following)



Joint attention = shared attention + the mutual knowledge of both individuals as being attentive towards the same external entity, <u>accomplished via communicative cues.</u>



Joint attention is essential for joint action and coordination + it itself can be a joint action (“Look!”)

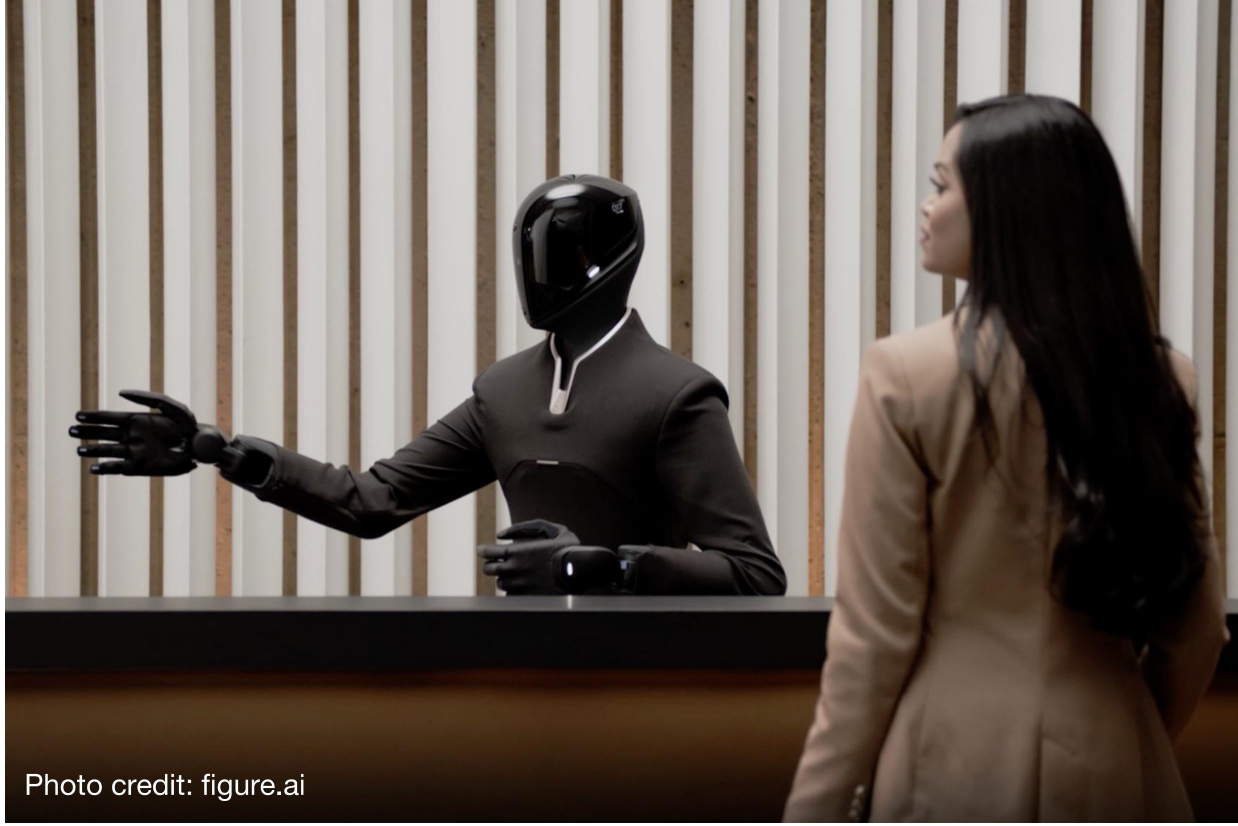


Photo credit: figure.ai

Fiebich, A., & Gallagher, S. (2013). Joint attention in joint action. Philosophical Psychology, 26(4), 571-587.

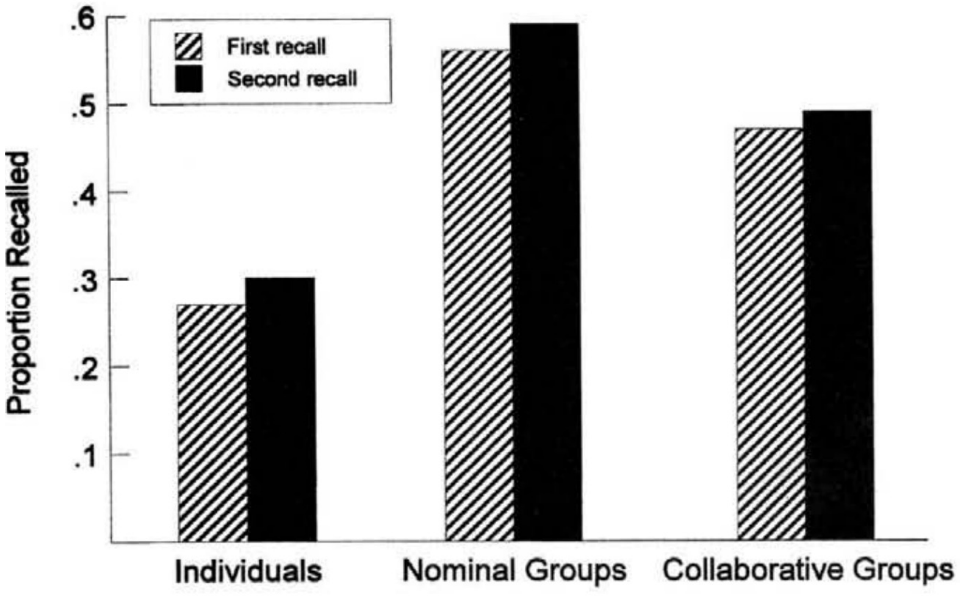
Higher-level phenomena: group memory, group decision-making, and conversational alignment

Group memory

Example: recalling the last project meeting together

People can collaborate to recall events, such that **group memory** = an emergent property of different individuals' recollections which are expressed within a social context.

Experimental setup: three people individually memorize words and pictures and then recall them together



An explanation — **collaborative inhibition**:

- Free-riding (motivational factor, less important because increasing motivation didn't eliminate the effect *)

- Individual retrieval-blocking (cognitive factor)

Weldon, M. S., & Bellinger, K. D. (1997). Collective memory: collaborative and individual processes in remembering. Journal of experimental psychology: Learning, memory, and cognition, 23(5), 1160.

* Weldon, M. S., Blair, C., & Huebsch, P. D. (2000). Group remembering: Does social loafing underlie collaborative inhibition?. Journal of Experimental Psychology: Learning, Memory, and Cognition, 26(6).

Group memory

The interests and characteristics of the group bias members' attention to and reconstruction of events.

(a) appropriateness in a particular social setting

(b) who speaks when and whose recollections receive the most weight

(c) there can be different purposes of the recollective activity: e.g., assuring mom and dad they have nothing to worry about vs. cultivating peer relationships

There is also a tendency to discuss info known by everyone else (known as **mutual enhancement**).

In the long run, this affects what we can recall later individually.

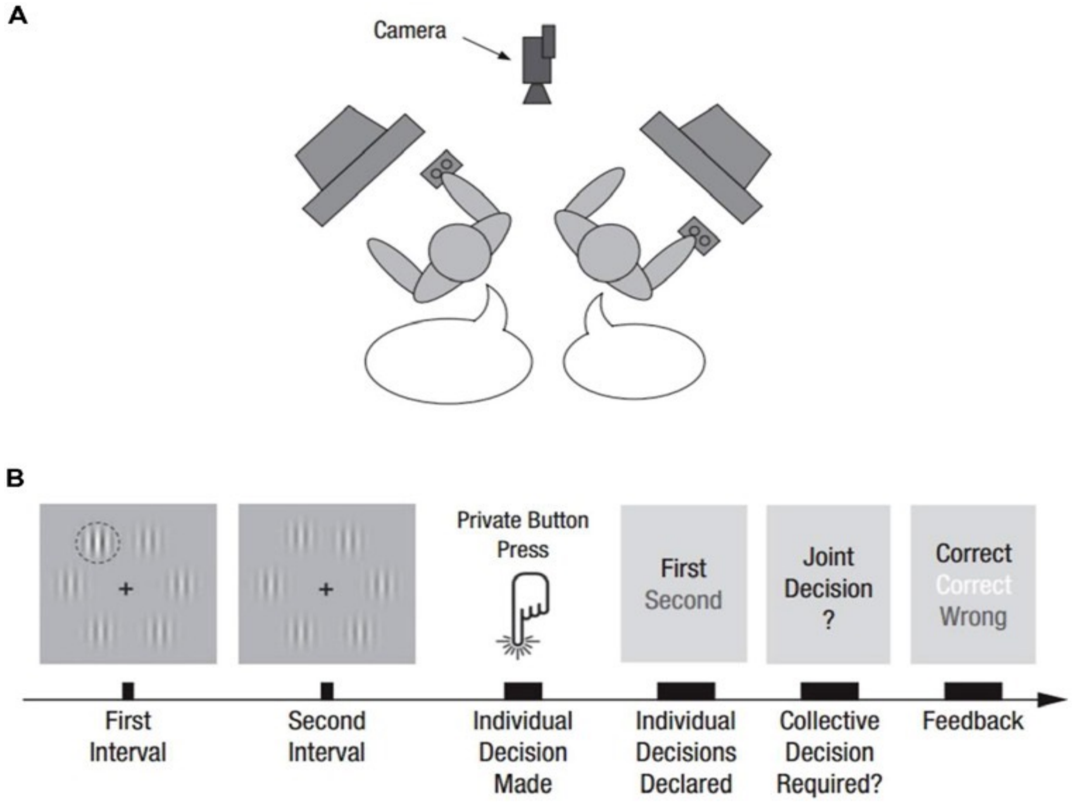
Weldon, M. S., & Bellinger, K. D. (1997). Collective memory: collaborative and individual processes in remembering. Journal of experimental psychology: Learning, memory, and cognition, 23(5), 1160.

Harris, C. B., Paterson, H. M., & Kemp, R. I. (2008). Collaborative recall and collective memory: What happens when we remember together?. Memory, 16(3), 213-230.

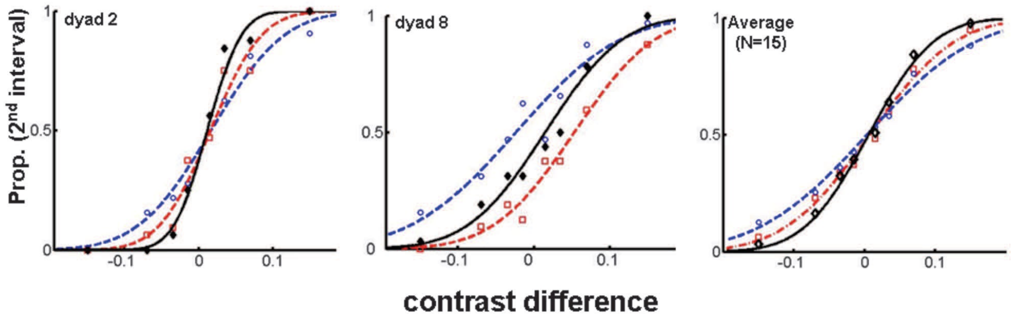
Wittenbaum, G. M., & Park, E. S. (2001). The collective preference for shared information. Current Directions in Psychological Science, 10(2), 70-73.

Group decision-making

Like with group memory, not always are two heads better than one (but sometimes!)



○ less sensitive observer, s_{min}
□ more sensitive observer, s_{max}
◇ dyad performance, s_{dyad}

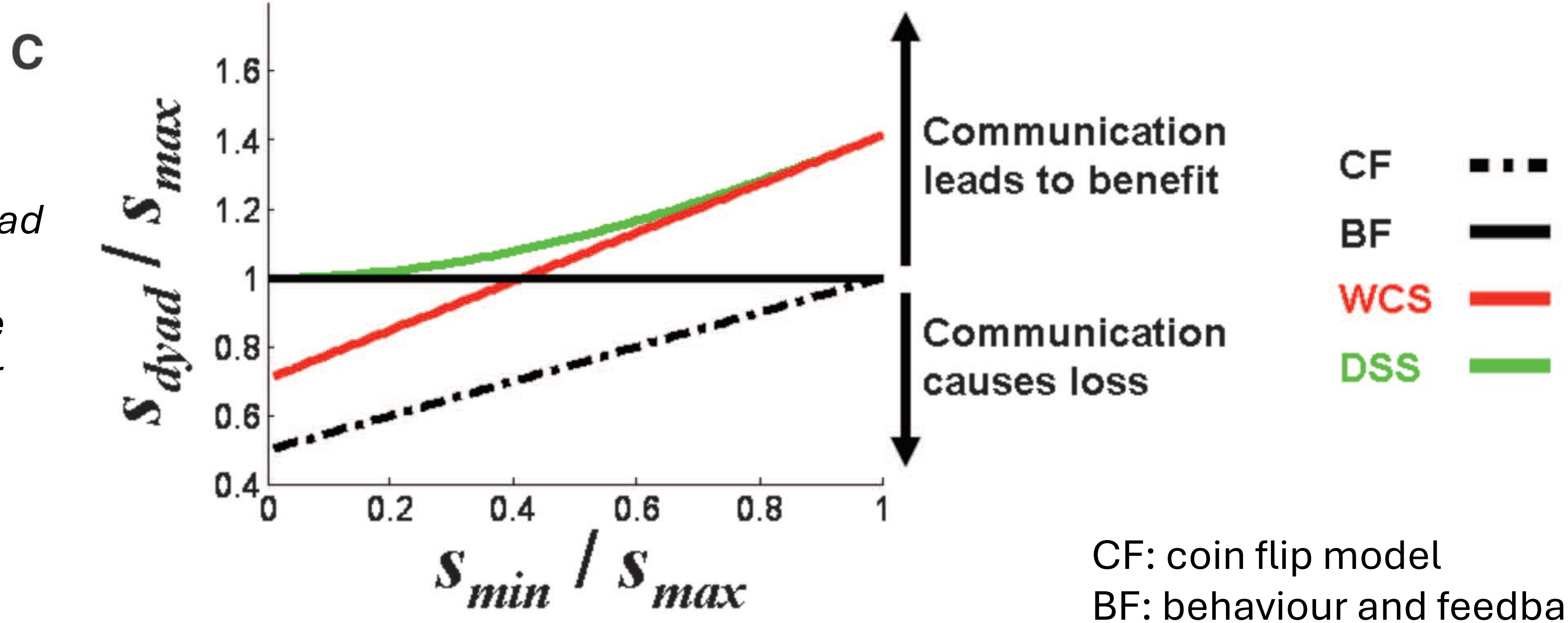


Bahrami, B., Olsen, K., Latham, P. E., Roepstorff, A., Rees, G., & Frith, C. D. (2010). Optimally interacting minds. Science, 329(5995), 1081-1085.

Group decision-making

Predictions of 4 models:

Ratio of dyad to more sensitive member



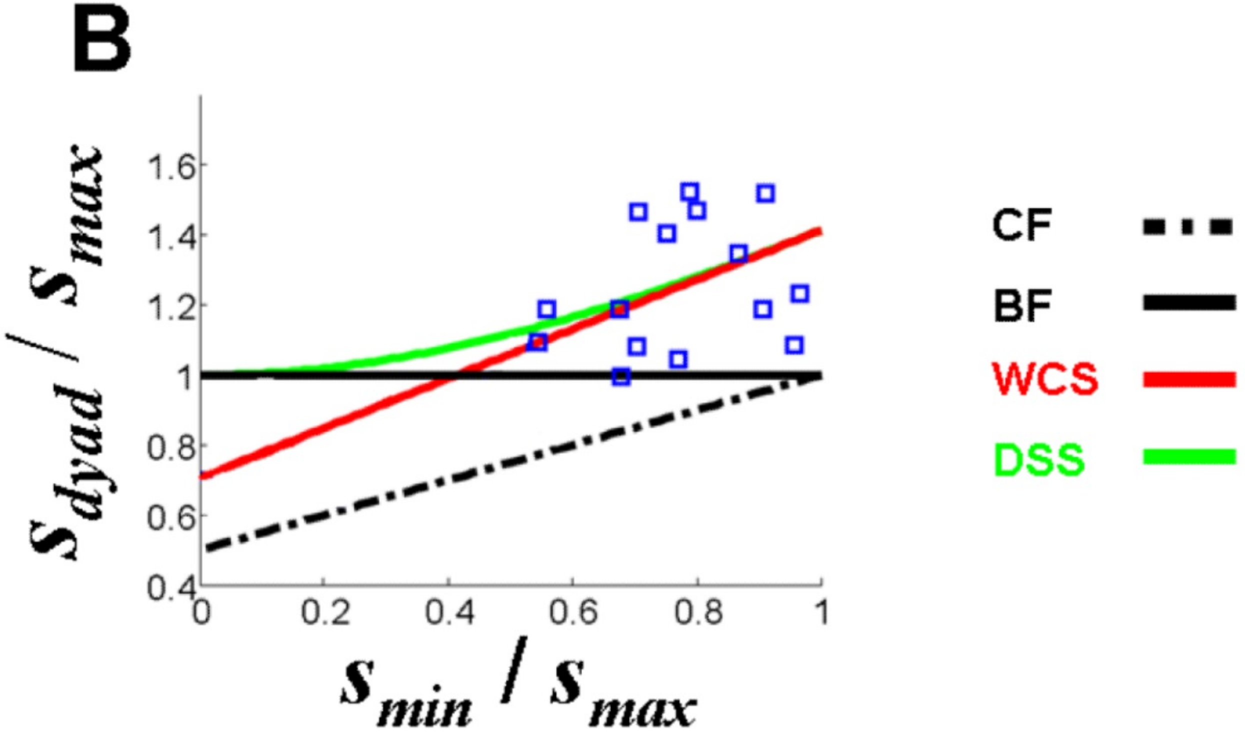
Ratio of individual sensitivities

CF: coin flip model
BF: behaviour and feedback model
WCS: weighted confidence sharing
DSS: direct signal sharing model

Bahrami, B., Olsen, K., Latham, P. E., Roepstorff, A., Rees, G., & Frith, C. D. (2010). Optimally interacting minds. Science, 329(5995), 1081-1085.

Group decision-making

When people have similar sensitivity, two heads are better than one, but we cannot see difference between the red and green models



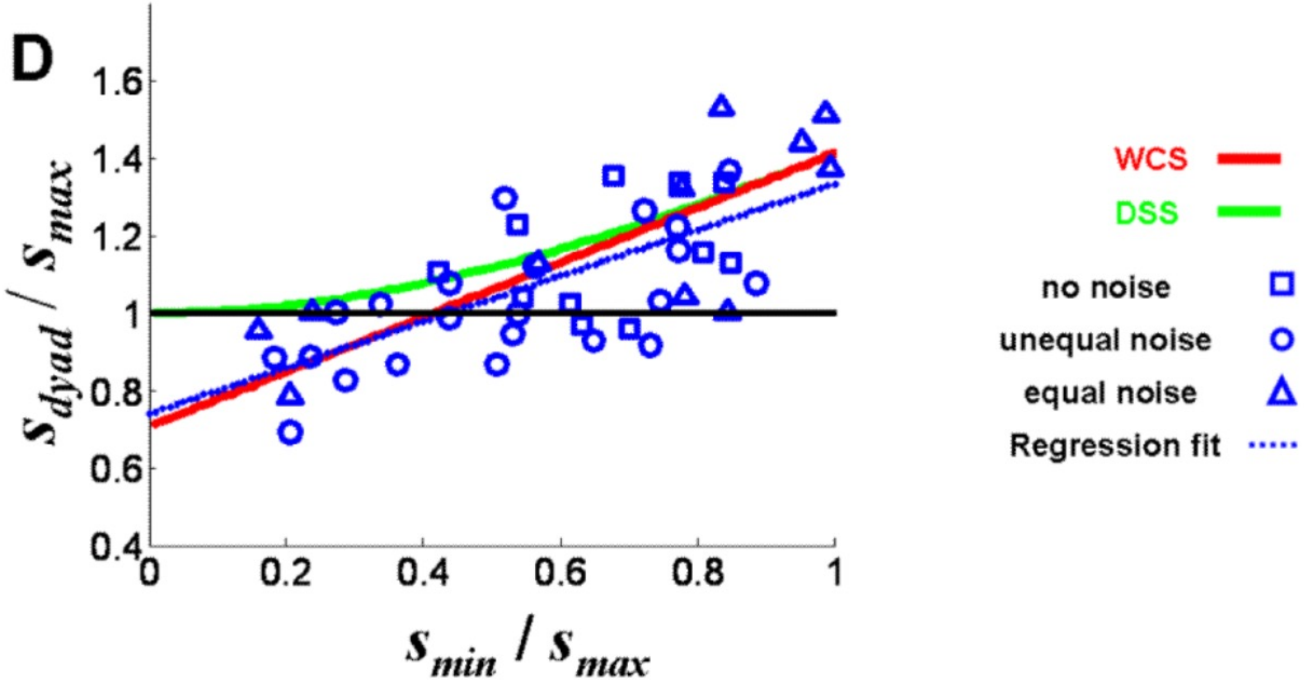
CF: coin flip model
BF: behaviour and feedback model
WCS: weighted confidence sharing
DSS: direct signal sharing model

Bahrami, B., Olsen, K., Latham, P. E., Roepstorff, A., Rees, G., & Frith, C. D. (2010). Optimally interacting minds. Science, 329(5995), 1081-1085.

Group decision-making

When people have different sensitivity, two heads can be worse than one.

Collective decision-making relies on Bayesian combination of confidence



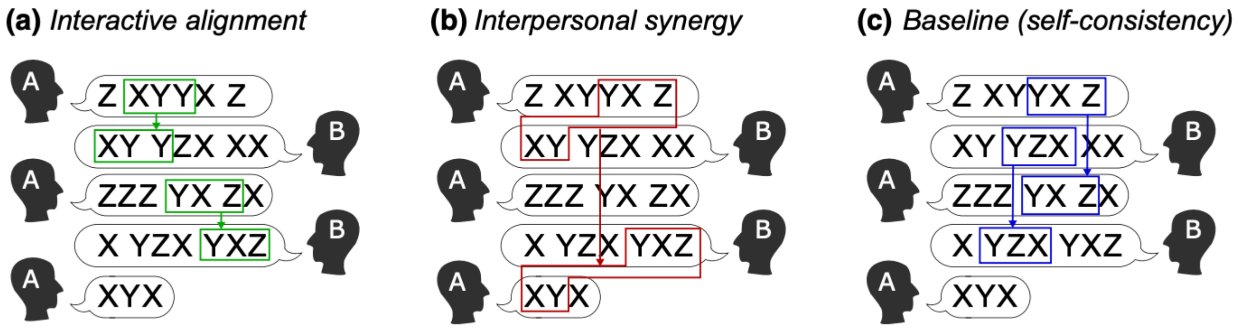
CF: coin flip model
BF: behaviour and feedback model
WCS: weighted confidence sharing
DSS: direct signal sharing model

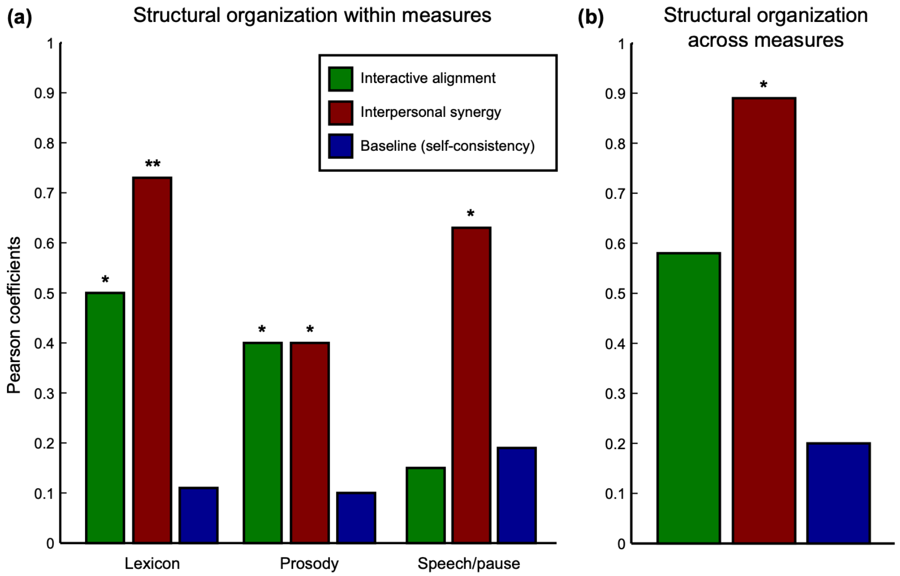
Bahrami, B., Olsen, K., Latham, P. E., Roepstorff, A., Rees, G., & Frith, C. D. (2010). Optimally interacting minds. Science, 329(5995), 1081-1085.

Conversational alignment

Interlocutors frequently adapt to each other's expressions — **interactive alignment**

Interdependence in the linguistic behavior of interlocutors, not limited to copying — **interpersonal synergy**
Example: highly typical question-answer structure of interaction with a cashier at a store

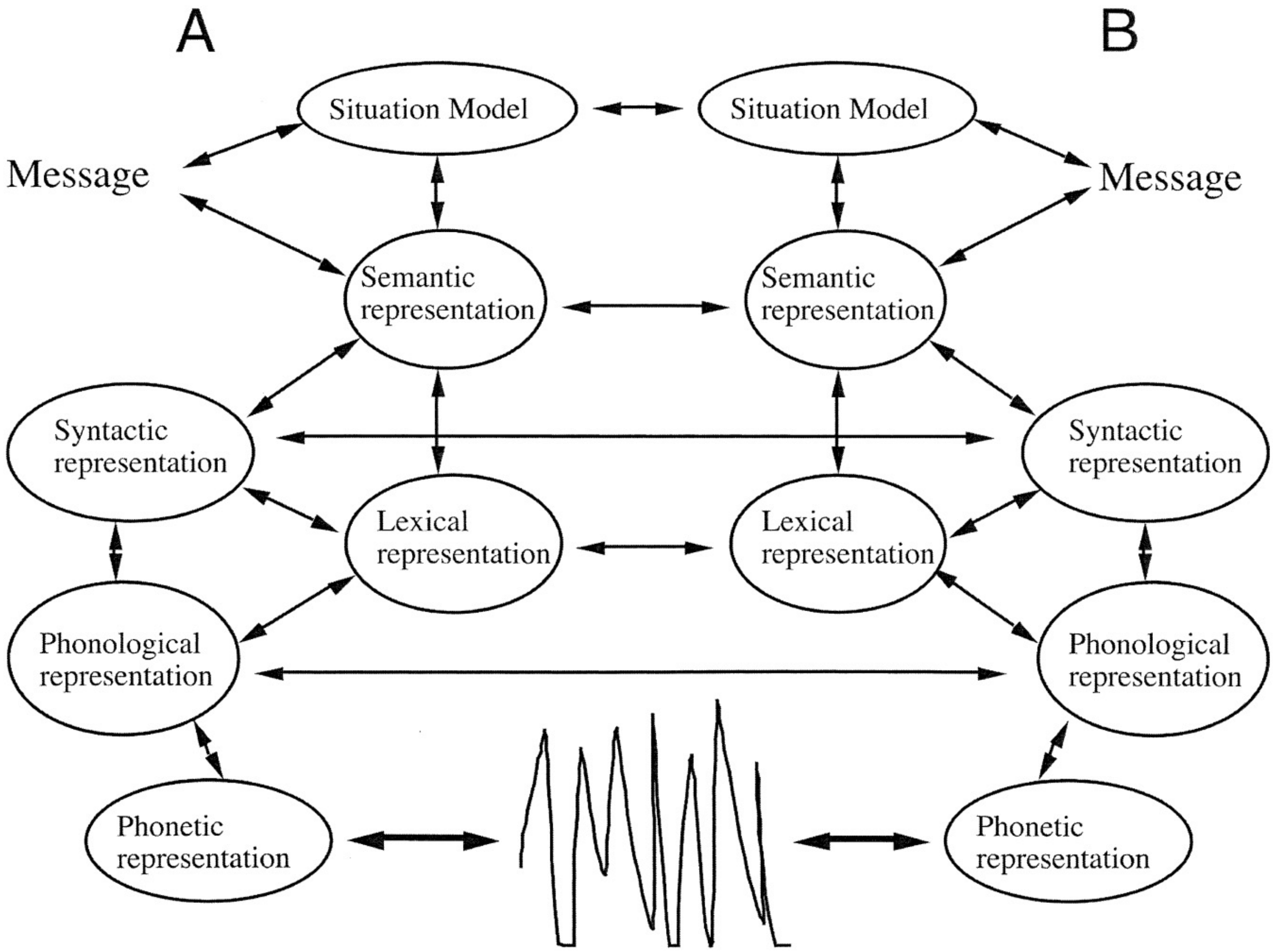




Both interactive alignment and interpersonal synergy were present more than by chance. However, interpersonal synergy model better predicted dyads' performance in the sensory decision-making task

Fusaroli, R., & Tylén, K. (2016). Investigating conversational dynamics: Interactive alignment, interpersonal synergy, and collective task performance. Cognitive science, 40(1), 145-171.

Conversational alignment



"Alignment at one level leads to alignment at another"

"Alignment of **situation models** is central to successful dialogue"

"A **situation model** is a multi-dimensional representation of the situation under discussion"

Pickering, M. J., & Garrod, S. (2004). Toward a mechanistic psychology of dialogue. Behavioral and brain sciences, 27(2), 169-190.

Conversational alignment

Alignment of situation models (or establishing the **common ground**) is stronger during interaction than during observation/comprehension

Three people:
- Director
- Matcher
- Overhearer

Don't see each other.

Director and matcher cat talk as much as they want.

Task for the Matcher: arrange the Tangram cards in the order as required by the Director

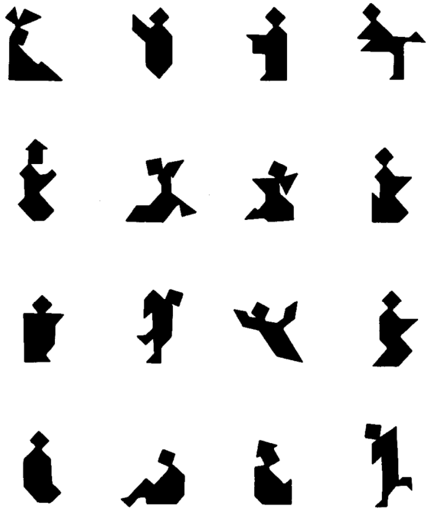
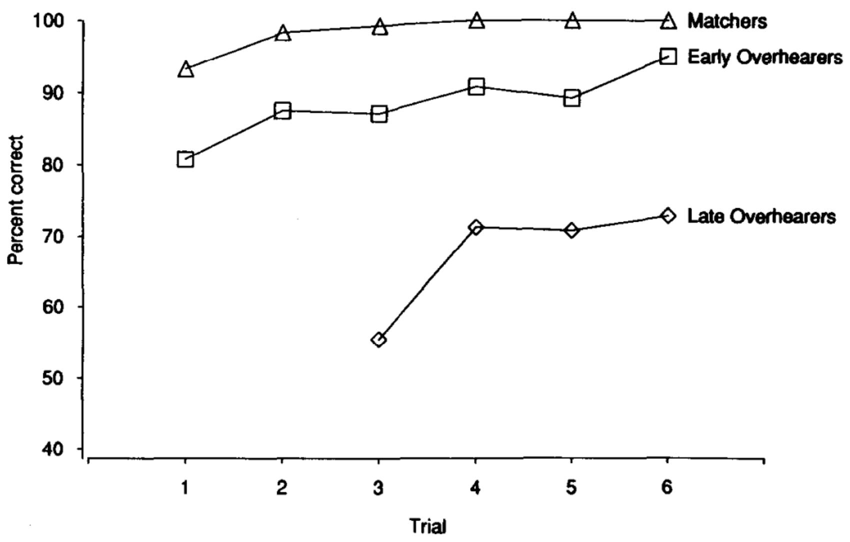


FIG. 1. The Tangram figures.

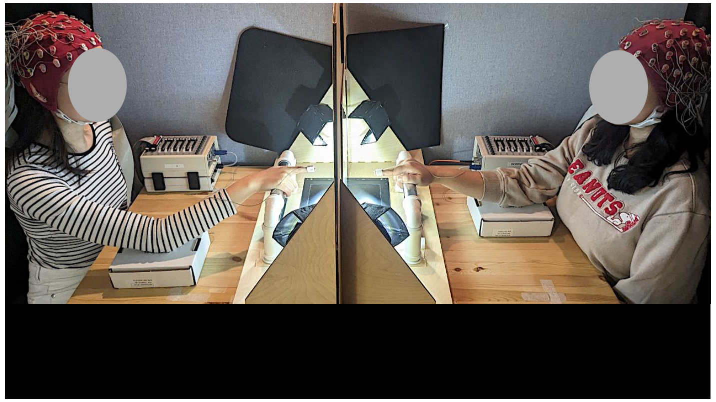


Schober, M. F., & Clark, H. H. (1989). Understanding by addressees and overhearers. Cognitive psychology, 21(2), 211-232.

How are these phenomena studied?

Group-based and realistic paradigms and methods

In the lab



more controlled

more naturalistic

In the "wild"

more controlled

natural

simultaneous recording of multiple brains at the same (popular in recent years) is called **hyperscanning**

Opportunities and limitations of the group-based approach to social cognition

	one-person approach	group-based approach
opportunities	• fine isolation of specific processes • less noise • simpler experimental paradigms • lab: faster • more measurable	• naturalistic • allows capturing unique phenomena • scalable: from pairs to thousands of performers
limitations	• ignores actual social interaction • unnatural, artificial	• quite noisy • many processes are intertwined • expensive • lab: slow • less-interpretable physiological and brain correlates

In sum, social interactions (and social cognition) rely on a combination of individual- and collective-level cognitive processes

Aliaksandr Dabranau, aldab@dtu.dk